



ADIDAS US SALES INTERACTIVE DASHBOARD



AI-Augmented Business Analytics with Python & Streamlit

Team: Team CJS (Cayden, Jules, Sophia)

Course: ISTM 635 – Python Programming | Pepperdine University | MSBA | Spring 2026

PROBLEM STATEMENT & OBJECTIVE



PROBLEM

Adidas US sales data (2020-2021) spanning **6 retailers, 5 regions, and 6 product categories** lacks unified real-time visibility for strategic decision-making.

OBJECTIVE

Build an interactive Python dashboard using Streamlit and Plotly to transform \$900M in sales data into **actionable business insights**.

** Note: Alternative dataset approved by instructor in class.*



\$899,902,125

TOTAL SALES



\$332,134,761

OPERATING PROFIT



2,478,861

UNITS SOLD



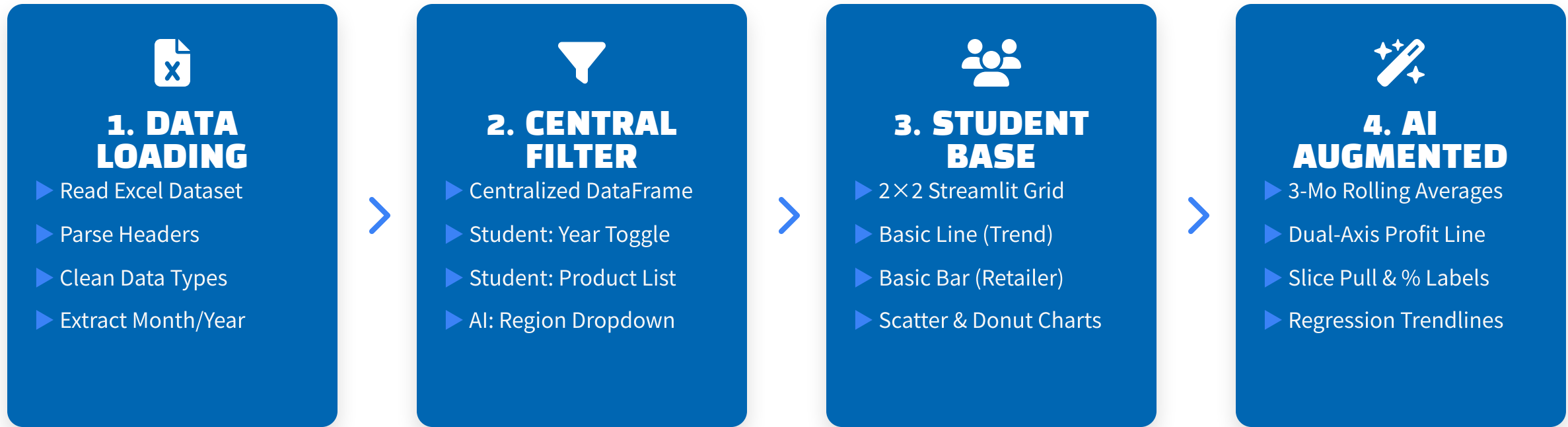
9,648

TRANSACTIONS

DESIGN ARCHITECTURE



TWO-TRACK ARCHITECTURE: STUDENT BASELINE VS. AI-AUGMENTED



ONE FILTERED DATAFRAME FEEDS ALL VISUALIZATIONS - TRUE INTERACTIVE LINKING

TECHNICAL IMPLEMENTATION



CORE STACK & CODE ARCHITECTURE



Python 3.x



Streamlit



Plotly / PX



Pandas/NumPy



GitHub Copilot

IMPLEMENTATION HIGHLIGHTS



@st.cache_data

Optimized data loading & caching



st.sidebar

Filters: Student (Year+Product), AI (Region)



st.columns

Responsive 2×2 dynamic grid layout



Real-time KPIs

Auto-updating metric tracking cards

dashboard_snippets.py

```
@st.cache_data
def load_data():
    df = pd.read_excel("Adidas_US_Sales.xlsx")
    df["Year"] = df["Invoice Date"].dt.year
    return df

df = load_data()

# Student Baseline Filters
sel_year = st.sidebar.radio("Year", ["Both", "2020", "2021"])
sel_prod = st.sidebar.multiselect("Product", products)
# AI-Augmented Filter
sel_reg = st.sidebar.selectbox("Region", regions)

# Centralized Filtering
filtered = df[(df["Year"] == sel_year) & (df["Region"] == sel_reg)]
```

VISUAL SHOWCASE 1 & 2



SALES TRENDS & RETAILER PERFORMANCE

VIZ 1: Monthly Sales Trend



 AI-Augmented

Added 3-month rolling average trendline (dashed orange) to smooth seasonal noise.

 Student Baseline

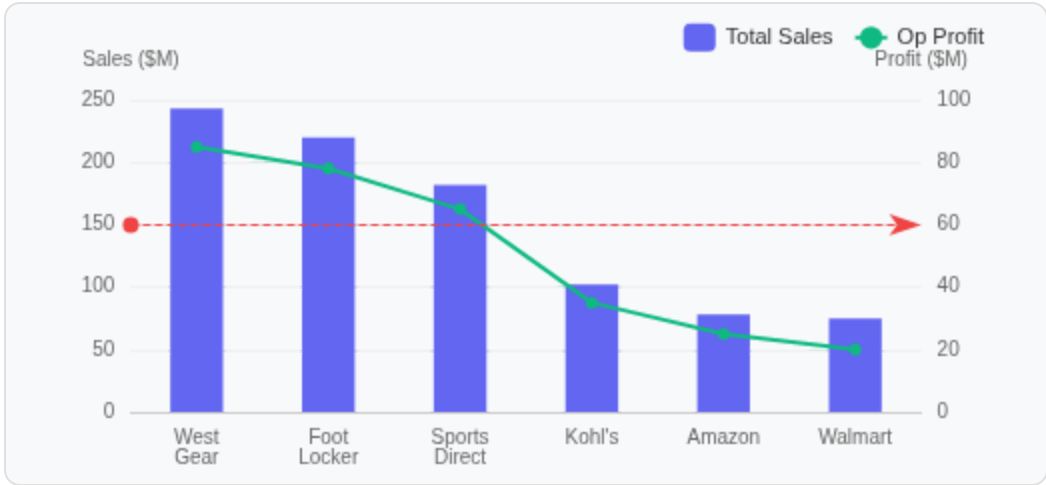
Basic line chart with markers, using blue color for clear data presentation.



Business Insight

Sales peaked mid-2021, showing strong upward momentum. The rolling average effectively smooths out short-term seasonal fluctuations.

VIZ 2: Total Sales by Retailer



 AI-Augmented

Overlaid operating profit on a secondary Y-axis and added an average sales reference line (\$150M).

 Student Baseline

Simple bar chart using purple color to visualize total sales by retailer.



Business Insight

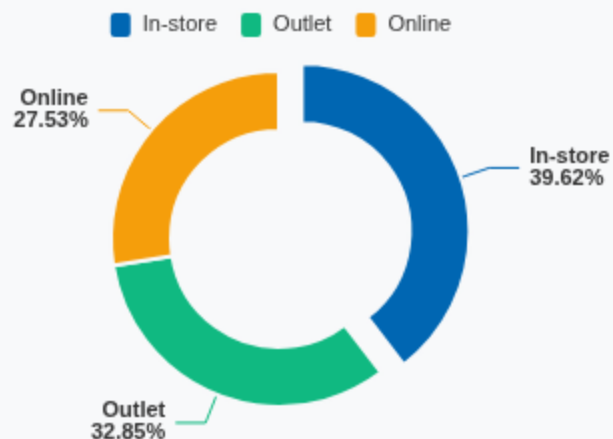
West Gear (\$243M) and Foot Locker (\$220M) account for 51% of total sales, but operating profit margins vary distinctly by retailer.

VISUAL SHOWCASE 3 & 4



SALES METHODS & PROFITABILITY ANALYSIS

VIZ 3: Sales by Channel Method



AI-Augmented

Added dynamic slice explosion (pulls largest slice by 8%) + percentage labels (textinfo='label+percent').

Student Baseline

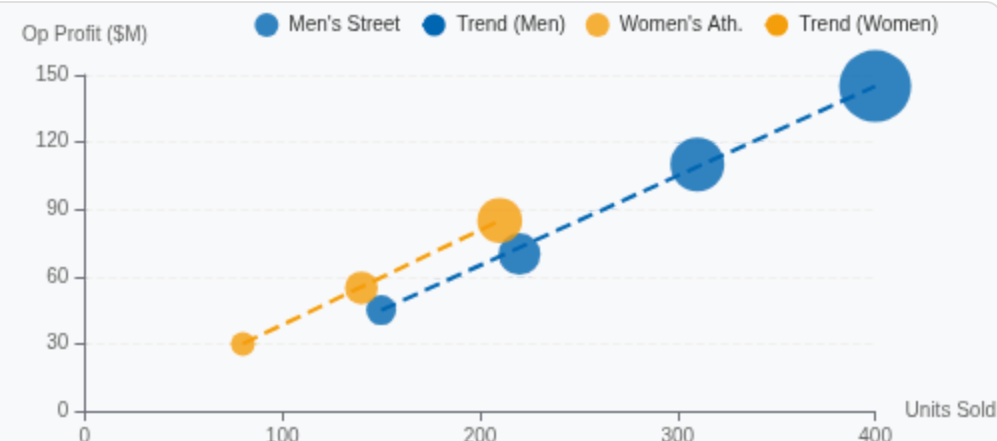
Basic donut chart with hole=0.4, default colors.



Business Insight

In-store dominates at 39.6% (\$357M), but online channel growing - opportunity for omnichannel strategy.

VIZ 4: Units Sold vs. Operating Profit



AI-Augmented

Added bubble size scaling (normalized by Total Sales), hover data enrichment (Retailer + Region), product-level regression trendlines using np.polyfit.

Student Baseline

Simple scatter plot colored by product.



Business Insight

Men's Street Footwear shows highest volume AND profitability; Women's Athletic has untapped potential.

AI VS HUMAN



TECHNICAL COMPARISON

Visualization	Student Baseline (student_dashboard.py)	AI-Augmented (dashboard.py)	Technical Analysis
Viz 1: Line Chart	Simple line + markers, blue color	3-month rolling avg calculation + dashed trendline	AI: +8 lines (rolling computation), increases analytical depth
Viz 2: Bar Chart	Simple bar chart, purple color	<code>make_subplots</code> dual-axis + profit overlay + avg reference line	AI: +15 lines (secondary axis logic), significantly more complex but richer insights
Viz 3: Donut Chart	Basic donut hole=0.4	Dynamic pull array + textinfo labels	AI: +6 lines (max detection + pull calculation), enhanced visual emphasis
Viz 4: Scatter Plot	Simple scatter by product	Bubble sizing + <code>hover_data</code> + <code>linear_regression</code> loop	AI: +12 lines (normalization + regression), major analytical upgrade



Summary Metrics

Student baseline provides a clean, maintainable foundation; AI augmentation adds significant analytical depth, averaging +10 lines of code per visualization.

AI VS HUMAN



BUSINESS VALUE COMPARISON

Visualization	AI-Augmented Value	Student Baseline Value	Business Impact Winner
Viz 1: Sales Trend	Rolling average reveals underlying trends, filters seasonal noise for forecasting	Clean baseline for stakeholder review	AI — enables trend-based forecasting
Viz 2: Retailer Performance	Profit overlay enables margin analysis across retailers	Simple visual for quick sales comparison	AI — supports profitability-driven decisions
Viz 3: Sales Methods	Exploded slice draws attention to dominant channel	Basic donut chart showing channel share	Student — actionable precision for channel budgeting
Viz 4: Profitability	Trendlines reveal product-level ROI patterns	Clean scatter enables drill-down exploration	AI — supports strategic product portfolio decisions



Conclusion: The Hybrid Advantage

AI excels at **analytical depth and pattern detection**; Student baseline excels at **clarity and precision**; the hybrid approach combining both creates maximum business value.

CHALLENGES & SOLUTIONS



OVERCOMING TECHNICAL & IMPLEMENTATION HURDLES

01. INTERACTIVE LINKING



The Challenge

Each chart was initially independent; applying a filter change only updated one visualization at a time, preventing a unified data view across the dashboard.



The Solution

Implemented **centralized DataFrame filtering**. Centralized sidebar filters update a shared filtered DataFrame that feeds all four visualizations.

02. STREAMLIT ENVIRONMENT



The Challenge

Google Colab is not ideal for running Streamlit apps, as Streamlit requires a local server and port forwarding which Colab limits.



The Solution

Switched development environment entirely to **VS Code**. This provided the perfect local hosting environment for seamless Streamlit testing and iteration.

03. AI ASSISTANCE BALANCE



The Challenge

GitHub Copilot suggestions sometimes conflicted with existing code structures or our core learning objectives for understanding the underlying data flow.



The Solution

Adopted a **modular code architecture**. Each visualization was isolated in its own code block, allowing selective adoption of AI suggestions to maintain clarity.

CONCLUSION & KEY TAKEAWAYS



TRANSFORMING \$900M OF RAW DATA INTO ACTIONABLE INTELLIGENCE

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This project demonstrates the power of **AI-human collaboration** in business analytics. By combining GitHub Copilot's analytical enhancements with student-authored baseline visualizations, Team CJS transformed \$900M of Adidas sales data into an interactive Streamlit dashboard. The student baseline (student_dashboard.py) established clean, maintainable visualizations with two powerful interactive filters (Year toggle and Product selector). The AI-augmented version (dashboard.py) added analytical depth through rolling averages, dual-axis charts, regression trendlines, and enhanced interactivity. The result: a unified dashboard that empowers decision-makers to explore sales trends, retailer performance, channel mix, and product profitability in real time.



+ Human > AI Alone

Copilot accelerated development, students ensured clarity and business relevance.



Interactive Linking is Essential

Centralized DataFrame filtering solved the core challenge elegantly.



Streamlit + Plotly = Rapid Deployment

From data to insight in hours, not weeks.